Name _____ _____ Baggett/Fall2026/207/Quiz1

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) The average rate of change of f on $[a, b]$ is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

True **False**

Why? **That is the definition:** $\frac{f(b)-f(a)}{b-a}$ **is exactly the secant slope.**

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2. (**3** pts) If $\lim_{x\to 2} f(x) = 5$, then $f(2) = 5$.

True **False**

Why? **Test** $f(x) = \frac{x^2-4}{x-2}$ **at** $a = 2$**: the limit is** 4**, but** $f(2)$ **is not even defined. The limit ignores the value at the point.**

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3. (**4** pts) Let $f(x) = 3 - x^2$. Form the difference quotient at $a = 2$ and simplify it as far as it will go:

$$\frac{f(2+h) - f(2)}{h}$$

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$f(2+h) = 3 - (2+h)^2 = 3 - 4 - 4h - h^2 = -1 - 4h - h^2, \quad f(2) = -1.$

$$\frac{(-1 - 4h - h^2) - (-1)}{h} = \frac{-4h - h^2}{h} = \mathbf{-4 - h}$$

4. (**4** pts) A drone rises straight up. Its height above the ground after t seconds is

$$s(t) = 40t - 5t^2 \text{ meters}, \qquad 0 \le t \le 8.$$

(a) Find the drone's **average velocity** over the interval $[1, 3]$. Include units.

$s(3) = 75$**,** $s(1) = 35$**, so** $\frac{75-35}{3-1} = 20$ **m/sec.**

(b) In **one sentence**, say what that number tells you about the drone.

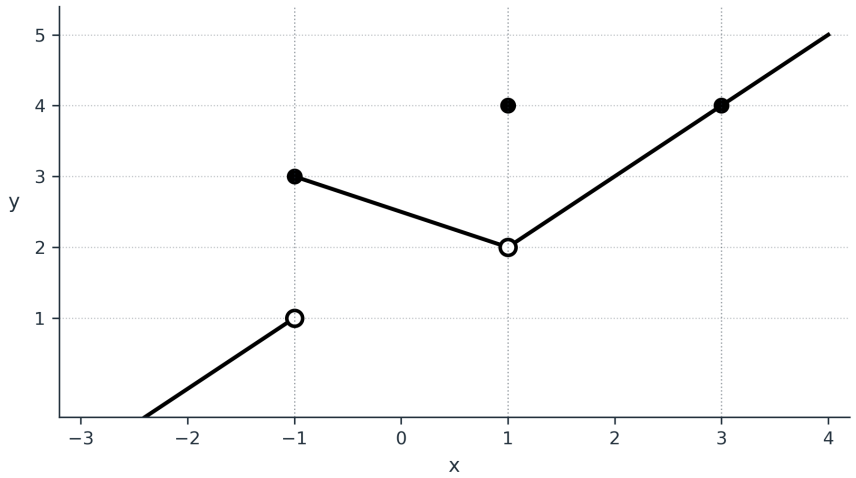
Over those two seconds the drone gained height at an average rate of 20 meters per second.

(c) Over $[1, 1.5]$, $[1, 1.1]$ and $[1, 1.01]$ the average velocities are 27.5, 29.5 and 29.95 m/sec. What number are they heading for, and what does it say about the drone at the **instant** $t = 1$?

30 **m/sec — the instantaneous velocity at** $t = 1$**.**

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5. (**6** pts) Use the graph of $y = f(x)$ below. If a limit does not exist, say so.



(a) $\lim_{x \to -1^-} f(x) =$ 1 (b) $\lim_{x \to -1} f(x) =$ **DNE**

(c) $\lim_{x \to 1} f(x) =$ 2 (d) $f(1) =$ 4

(e) $\lim_{x \to 3} f(x) =$ 4 (f) $f(3) =$ 4

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